Day 5: Active Reconnaissance & Nmap Scanning

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1. Introduction

After gathering public information in Day 4, it was time to "touch" the target. Today, we moved into **Active Reconnaissance**. This phase involves interacting directly with the target machine to see which ports are open, which services are running, and what operating system is being used.

2. Objective of the Day

The goal of Day 5 is to master **Nmap**, the industry-standard tool for network discovery. I focused on identifying open "doors" (ports) on the Metasploitable2 machine and understanding how to read scan results to find potential vulnerabilities.

3. Understanding Nmap (Network Mapper)

The Three States of a Port:

- **Open:** The service is actively listening for connections (e.g., a web server on port 80).
- **Closed:** The port is reachable but no service is running on it.
- **Filtered:** A firewall is blocking Nmap from determining if the port is open or closed.

4. Types of Scans

A. Stealth Scan (-sS) vs. TCP Connect (-sT)

- **TCP Connect Scan (-sT):** Completes the full "Three-Way Handshake." It is very accurate but very easy for firewalls to log.
- **SYN Stealth Scan (-sS):** This is the default. It sends a SYN packet but breaks the connection before it's finished. This often avoids triggering basic logs.

B. Version & OS Detection

- **-sV:** Probes open ports to determine what software and **version** is running.
- **-O:** Attempts to guess the **Operating System** (Linux, Windows, etc.) by analyzing how the stack responds to packets.

5. What I Did Today

- **Host Discovery:** I used `nmap -sn [Network/24]` to find all live devices in my virtual lab.
- **Service Enumeration:** I ran a version scan (`-sV`) on the Metasploitable2 IP.
- **Aggressive Scanning:** I practiced using the `-A` flag, which combines OS detection, version detection, script scanning, and traceroute in one command.
- **Output Management:** I learned how to save my results into a file using `-oN` (Normal format) so I can review them later without re-scanning.

6. What I Learned

- **Service Vulnerability:** I learned that an open port isn't a vulnerability by itself, but the **version** of the service running on it might be (e.g., an old version of FTP).
- **Timing is Everything:** Nmap has timing templates (-T0 to -T5). I learned that -T4 is great for lab environments, while -T1 or -T2 is used to stay "sneaky" on real networks.
- **The Power of Scripts:** I discovered the Nmap Scripting Engine (NSE), which allows Nmap to do more than just scan it can check for specific bugs automatically.

Key Takeaway for Newbies:

"Nmap is the eyes of a hacker. Without it, you are attacking in the dark."

7. Hands-on Examples: Scanning the Lab

- **Task:** Perform a full service and OS detection scan on the victim machine.

Execution:

1. `nmap -sV -O [Victim-IP]`
2. Check the "STATE" and "SERVICE" columns.
3. Identify the version of the "vsftpd" service.

Screenshot 1: Your terminal showing the output of an Nmap scan (e.g., `nmap -sV [IP]`).

```
┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ nmap -sV -O [redacted]
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 03:06 -0400
Nmap scan report for [redacted]
Host is up (0.00092s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: [redacted] (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.74 seconds
```

Screenshot 2: A screenshot showing the OS detection results (e.g., "Running: Linux 2.6.X").

```
Device type: general purpose
Running: Linux 2.6.X
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